When Finite Differences Flip the Sign of Second-Order Generative Geometry: A Non-Recoverability Result and an Exact Instrument

Anonymous authors
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Abstract

Measuring the curvature of a generative model's image manifold by finite differences can return the wrong sign. On Stable Diffusion, the rank correlation between a Jacobian singular value and a second-order curvature ratio is -0.73 under exact forward-mode automatic differentiation and a sign-unstable -0.18 under a finite-difference estimate of the identical quantity. The same finite differences recover the first-order Jacobian accurately. The cause is an asymmetry. A central finite difference is well conditioned for a first derivative but *non-recoverable* for a second, because truncation error and floating-point cancellation cannot be made small at the same step. The resulting relative-error floor, of order $\sqrt{\varepsilon_{\text{mach}}}$, is large in single precision. On a StyleGAN generator the naive second-difference curvature misses the exact value by at least 45% at every fixed step, in fp32 and fp64 alike, while the first-order difference of the same code agrees with exact AD to 2.4%. We give the exact alternative, composed forward-mode AD, which returns the k-th-order pushforward in k forward passes at the memory of one. An audit of how the latent-geometry literature computes second-order quantities finds the $/\varepsilon^2$ estimate to be the natural choice, yet one that most published methods sidestep by construction. We do not claim any published result is wrong. We characterize when finite differences fail for second-order generative geometry, give the first/second-order asymmetry as practitioner guidance, and supply a drop-in exact replacement.

1 Introduction

Let $G : \mathbb{R}^d \to \mathbb{R}^{3HW}$ be a trained generator. Treating the image set $\{G(z)\}$ as a manifold pulled back through G, a line of work studies its geometry: the pullback metric $M_z = J_z^\top J_z$ and its singular vectors (the local latent basis), geodesics between latent codes, and the curvature of the generated manifold (Arvanitidis et al., 2018; Shao et al., 2018; Chen et al., 2018; Park et al., 2023; Saito & Matsubara, 2025). The first-order objects J_z and M_z are routine to compute. The second-order objects are not: the curvature, the second-order pushforward $\partial^2 G/\partial\alpha^2$ along a path, and the metric derivative $\partial_z M_z$ that appears in the geodesic ODE. These are where the numerical method can silently flip a result.

We start with a measurement (Section 2). A natural second-order diagnostic on Stable Diffusion returns the opposite sign depending only on whether the second derivative is taken by finite differences or by exact forward-mode AD. The exact method finds a clean, seed-consistent relationship. Finite differences, applied to the same quantity, are sign-unstable and wash it out. A practitioner who reached for the natural finite-difference estimate would draw the wrong qualitative conclusion.

The explanation is an asymmetry (Section 3). A central finite difference is well conditioned for a first derivative but non-recoverable for a second, because for the second derivative truncation and floating-point cancellation trade off and cannot be small together. The resulting relative-error floor is of order $\sqrt{\varepsilon_{\text{mach}}}$. In the single precision that GANs and diffusion U-Nets run in, that floor is large, and on a strongly nonlinear generator it is larger still: at least 45% on StyleGAN, in both single and double precision (Section 4).

We contribute the reproducible sign-inversion phenomenon and the first/second-order asymmetry that governs it; an exact, constant-memory instrument for the higher-order pushforward, built by composing forward-mode AD; and an audit that locates which parts of the literature are exposed and which are safe. The contribution is a precise account of a real numerical hazard and the fix, not a claim that any published method is wrong; the audit finds that most route around the regime by construction (Section 7).

2 The sign-inversion phenomenon

Setup. Throughout, the exact instrument is forward-mode automatic differentiation composed with itself: nesting two Jacobian–vector products (JVPs) returns the second-order directional derivative $\partial_\alpha^2 G$ exactly, with no step size and at the memory of one forward pass (Section 5 gives the construction). The finite-difference baseline replaces the inner derivative by a central difference of the JVP at step ε. We run both on Stable Diffusion v1.5 in h-space. From a base latent we form the first-order Jacobian J of the truncated DDIM denoising map along K=16 random h-space probes, take its top k=12 right-singular directions u_i, and along each measure a second-order curvature ratio

$$\rho_i \;=\; \frac{|\,\partial_\alpha^2 G(z+\alpha u_i)\,|}{|\,\partial_\alpha G(z+\alpha u_i)\,|}\Bigg|_{\alpha=0},$$

the magnitude of the second-order pushforward relative to the first.

We then ask a standard question: does curvature co-vary with the first-order singular value σ_i? We compute Spearman(σ,ρ) with the second derivative in ρ taken both ways, exact and finite-difference.

Over 6 seeds, with the top 12 directions each, the two methods disagree in sign (Table 1). The exact instrument gives a strong negative correlation on five of six seeds, clustered in $[-0.94,-0.82]$, with seed 4 a near-zero outlier at -0.03; the mean is -0.73. Run through finite differences, the same ratio is sign-unstable across seeds and averages -0.18. The finite difference does not blur the relationship; it erases its sign. The exact estimate is more negative than the finite-difference estimate on all six seeds (Wilcoxon signed-rank $p=0.031$, the smallest value attainable at n=6), and the finite-difference estimate is sign-unstable at every step in the grid $\{0.1, 0.05, 0.01, 0.001\}$, with means between -0.08 and -0.34, so no step reproduces the clean -0.73. A study built on the finite-difference route would report “no clear relationship,” the opposite of what the exact second-order estimate finds.

seed	0	1	2	3	4	5	mean
exact composed JVP	-0.85	-0.83	-0.92	-0.94	-0.03	-0.84	-0.73
FD-of-JVP(ε=0.05)	$+0.18$	-0.55	-0.78	-0.40	$+0.55$	-0.07	-0.18

Table 1: Spearman(σ,ρ) on Stable Diffusion, per seed. The exact second-order estimate is consistently negative; the finite-difference estimate of the identical quantity changes sign across seeds and averages near zero. We define ρ to expose the method dependence; the point is that the two numerical routes disagree on its sign, not the specific value of any prior work.

3 Why finite differences are non-recoverable for a second derivative

Proposition 1 (Second-order non-recoverability)**.** *Let $f \in C^4$ near α with $f^{(4)}$ bounded and $f''(\alpha) \neq 0$, evaluated in floating point with unit roundoff $\varepsilon_{\text{mach}}$. The central second difference $D_\varepsilon^2 f = \big(f(\alpha+\varepsilon)-2f(\alpha)+f(\alpha-\varepsilon)\big)/\varepsilon^2$ has relative error, minimized over the step,*

$$\min_{\varepsilon}\ \frac{|D_\varepsilon^2 f - f''(\alpha)|}{|f''(\alpha)|} = \frac{2\sqrt{|f^{(4)}(\alpha)\, f(\alpha)|}}{|f''(\alpha)|}\,\varepsilon_{\text{mach}}^{1/2} + o\big(\varepsilon_{\text{mach}}^{1/2}\big),$$

attained at $\varepsilon^\star \sim \varepsilon_{\text{mach}}^{1/4}$, and no step does better. The central first difference, by contrast, attains floor $\varepsilon_{\text{mach}}^{2/3}$.

The argument is elementary; we give it because the consequence is easy to miss in practice. The central second difference satisfies

$$D_\varepsilon^2 f - f''(\alpha) = \underbrace{\tfrac{\varepsilon^2}{12} f^{(4)}(\alpha) + O(\varepsilon^4)}_{\text{truncation}} + \underbrace{O\big(\varepsilon_{\text{mach}}\,|f|/\varepsilon^2\big)}_{\text{cancellation}},$$

and the two terms cannot be small together: minimizing their sum gives $\varepsilon^\star \sim \varepsilon_{\text{mach}}^{1/4}$ and the floor above (full derivation in Appendix C). The exponent $\frac{1}{2}$ is intrinsic to the operation; the constant carries the generator's nonlinearity $|f^{(4)}f|/f''^2$, which is large for a deep network. In double precision the floor is about 10^{-8} in the benign (constant near one) case. In the single precision generators use ($\varepsilon_{\text{mach}} \approx 1.2 \times 10^{-7}$ for fp32) it is about 3×10^{-4} benign, and far worse for a strongly nonlinear G, where Section 4 measures $\geq 45\%$. The floor is a property of the operation in fixed precision, not of any architecture, so the hazard is general; we demonstrate its magnitude on two generator families.

The first derivative behaves differently. The central first difference $\big(f(\alpha+\varepsilon) - f(\alpha-\varepsilon)\big)/2\varepsilon$ has truncation $O(\varepsilon^2)$ and cancellation $O(\varepsilon_{\text{mach}}|f|/\varepsilon)$, a much milder floor $\varepsilon_{\text{mach}}^{2/3}$ with a wide usable window. First-order Jacobians are therefore safe to finite-difference while second-order curvature is not. This is also the control we use throughout: wherever the first-order finite difference agrees with the exact JVP, the pipeline is correct, and any second-order disagreement is real rather than a bug.

To check the constants against ground truth, let $f(\alpha) = [\,e^{0.7\alpha}, \sin 3\alpha, (\alpha+0.4)^3, \tanh 2\alpha, \alpha^2 \cos\alpha\,]$, whose derivatives are known in closed form. At $\alpha = 0.3$ in double precision (Table 2) the central second-order relative error is U-shaped: it bottoms at 8.6×10^{-9} near $\varepsilon = 10^{-4}$ and diverges to 0.29 by $\varepsilon = 10^{-8}$. The composed JVP reaches 2.5×10^{-17}, so the best finite difference is 3.5×10^8 times worse.

ε	central FD 2nd-order rel. err.	
10^{-1}	7.9×10^{-3}	(truncation)
10^{-2}	8.0×10^{-5}	
10^{-4}	8.6×10^{-9}	(best)
10^{-6}	3.0×10^{-5}	
10^{-8}	2.9×10^{-1}	(cancellation)
composed JVP	2.5×10^{-17}	(exact)

Table 2: Second-order accuracy on the analytic toy at $\alpha = 0.3$, double precision. Finite differences never approach the exact composed JVP.

4 Quantification on a real generator

On a real generator the failure is worse than the toy predicts, and for a second reason. We compare the naive second-difference curvature a practitioner would write, $D_\delta^2 G = \big(G(z+\delta u) - 2G(z) + G(z-\delta u)\big)/\delta^2$, against the exact composed JVP $\partial_\alpha^2 G(z+\alpha u)$ on a StyleGAN-bedroom generator at 256×256, over 6 directions (3 annotated attributes, 3 random) and 16 samples, sweeping δ (Figure 1, Table 3). The exact curvature is a definite, step-free number, mean 1.17. Holding a single step across all samples, as a practitioner does, the finite-difference mean relative error never drops below 45.2% (best step $\delta = 3$, bracketed on both sides); it is 267% at $\delta \approx 0.3$ and explodes as $\delta \to 0$. Allowing an oracle per-sample step the median best error is still about 9%, against the composed JVP's zero.

Precision does not rescue it. Rerunning the entire sweep in fp64 leaves the floor unchanged, 47.1% in both fp32 and fp64, and the mechanism is not cancellation alone. At usable steps the error is truncation and is precision-blind. At tiny steps fp32 adds catastrophic cancellation, but fp64, which removes that, instead exposes the generator's non-smoothness: across a LeakyReLU kink the chord-scale second difference does not converge to the pointwise pushforward, and the fp64 error grows past 3000% as $\delta \to 0$. So no step and no precision recovers the exact second-order pushforward, which the composed JVP returns directly.

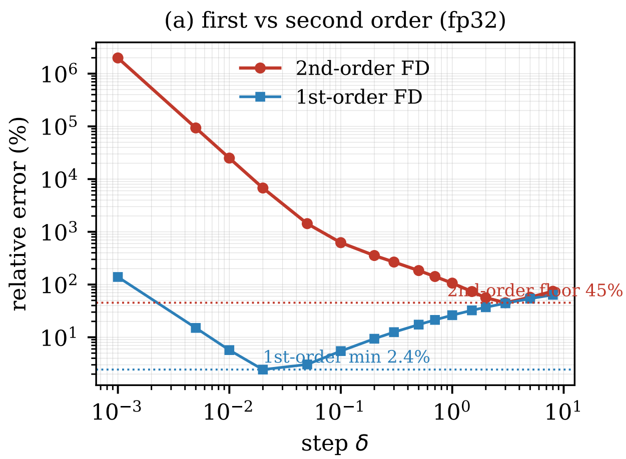

Figure 1: The non-recoverability of Proposition 1 on a StyleGAN generator. (a) Relative error of the finite-difference curvature versus step δ in fp32: the first-order difference (blue) has a usable minimum near 2.4%, while the second-order difference (red) never falls below 45%. The exact composed JVP has error 0 at any δ. (b) The second-order error in fp32 versus fp64: double precision removes the small-step cancellation but leaves the best-step floor unchanged at 47%, so no precision recovers the curvature.

The control confirms the instrument is correct. The first-order central difference of the same pipeline agrees with the exact first-order JVP to 2.4%, so the second-order failure is genuine.

step δ	$\lvert D^2_\delta G \rvert$	2nd-order rel. err.	1st-order rel. err.
8	7.1×10^{-3}	74.5%	63.8%
3	2.7×10^{-2}	45.2% (floor)	44.1%
1	9.9×10^{-2}	106.4%	26.2%
0.3	3.7×10^{-1}	266.9%	12.5%
0.1	1.06	622.8%	5.5%
0.02	4.05	6.8×10^3%	2.4% (1st-order min)
10^{-3}	4.09×10^2	2.0×10^6%	139%
composed JVP	1.17	exact (step-free)	—

Table 3: StyleGAN-bedroom, fp32. Columns are means over 6 directions $\times 16$ samples at each fixed step. The relative-error column is the mean per-direction relative error; $|D^2_\delta G|$ is the aggregate magnitude, shown for scale and not the quantity the relative error is computed from. The naive second difference never gets within 45% of the exact curvature at any fixed step, while the first-order difference of the same code agrees to 2.4%. Rerunning in fp64 leaves the best-step floor unchanged (47.1% fp32 vs. 47.1% fp64): the floor is truncation at usable steps and generator non-smoothness at tiny steps, not a single-precision artifact.

5 The exact instrument

Forward-mode AD evaluates, for a map f and tangent v, the Jacobian–vector product $\partial f(x)[v]$ alongside $f(x)$ in one pass, without forming the Jacobian. With $f = G$, $x = \gamma(\alpha)$, and $v = u$ this gives the exact first-order pushforward. The second-order term is the directional derivative of the first, so composing forward mode with itself,

$$\partial_\alpha^2 G(\gamma(\alpha)) = \partial_\alpha \big[\partial G(\gamma(\alpha))[u]\big][u],$$

is exact and is evaluated by nesting two JVP calls: in PyTorch, `torch.func.jvp` of a function that itself returns a JVP. No Jacobian is stored, so memory stays at one forward pass, and the same nesting yields any order in a constant number of passes. Composing forward-mode passes for higher-order directional derivatives is standard algorithmic differentiation (Griewank & Walther, 2008); the contribution here is to apply it as the exact ground truth against which the finite-difference practice fails. The instrument is generator-agnostic: we ran it unmodified on StyleGAN1 and 2 and on a latent-diffusion U-Net. Unlike reverse-mode Hessian materialization, whose memory is $O(3HW)$, its cost is independent of output dimension.

6 Related work and exposure audit

The finite-difference step-size tradeoff is classical numerical analysis (Press et al., 2007), and higher-order forward-mode AD is standard (Griewank & Walther, 2008); what has been missing is an account of where the tradeoff bites in the differential geometry of generative models, and how often. We include the papers we could identify that compute, or deliberately avoid, a second-order or geodesic quantity in a generative-model latent space, and classify each by how it obtains that quantity (the per-paper test is in Appendix G). Table 4 gives the result. The explicit $/\varepsilon^2$ second-difference of Section 3 is the obvious estimate for a practitioner adding a curvature measurement, but most published methods never form it. They keep the metric first order, replace the geodesic ODE with a discretized energy whose exact gradient is a well-conditioned discrete Laplacian rather than a $/\varepsilon^2$ curvature, use analytic Christoffel symbols, or differentiate the energy with autodiff (Yang et al., 2018); the first-order constructions (Park et al., 2023; Chen et al., 2018; Kwon et al., 2023) never take a second derivative. The one exposed published practice is the first-order finite difference of a network derivative, such as a score-Jacobian–vector product, which sits on the well-conditioned $\varepsilon_{\text{mach}}^{2/3}$ side of Section 3.

The field is mostly safe because the canonical formulations never form the estimate, but that safety is a property of those formulations rather than a law. As curvature- and geodesic-level analysis of diffusion models grows, for instance the energy-and-score geometry of Saito & Matsubara (2025) or any second-order extension of the first-order bases of Park et al. (2023), the $/\varepsilon^2$ estimate becomes the obvious next step. Table 4, the asymmetry, and the exact instrument are meant as a reference for that emerging work.

Paper	2nd-order quantity	how obtained
Shao et al. (2018)	discretized geodesic energy	exact discrete-energy gradient (Laplacian), coarse T; no $/\varepsilon^2$
Saito & Matsubara (2025)	geodesic (energy)	first-order score-differences $\|s(x_{i+1})-s(x_i)\|^2$; no $/\varepsilon^2$
Arvanitidis et al. (2018)	geodesic ODE (Christoffel)	analytic ∂M, numeric ODE integ.
Yang et al. (2018)	geodesic energy	autodiff
Park et al. (2023)	none (first-order basis)	autodiff power iteration
Chen et al. (2018)	none (first-order metric)	—
Kwon et al. (2023)	none (first-order)	CLIP-grad + MLP

Table 4: How representative latent-geometry papers obtain second-order quantities. The explicit $/\varepsilon^2$ second difference (Sections 3, 4) is largely avoided in published work, but it is the obvious choice for a practitioner adding a curvature measurement, which is where the hazard bites.

Guidance. Finite-difference first-order quantities freely: choose a step near $\varepsilon_{\text{mach}}^{1/3}$ and expect a few digits. Never finite-difference a second derivative. In single precision no step gives even one reliable digit on a deep generator (Section 4), and the error can flip the sign of a downstream conclusion (Section 2). Use composed forward-mode AD: it is exact, costs one extra forward pass per order, and adds no memory. It also gives a free check. If the first-order finite difference and the JVP disagree, the pipeline is wrong; if they agree, any second-order disagreement is real.

7 Scope and limitations

One limitation matters more than the rest. We show that the measurement can flip, not that any published number is wrong; the audit's point is that the field mostly avoids the trap. The sign inversion of Section 2 is on a ratio ρ that we define to expose method dependence, not a quantity from prior work. The remaining caveats are routine. The non-recoverability result is specific to second-order finite differences; first-order finite differences, including score-JVPs, are well conditioned, as we state. The instrument is algorithmically standard but operationally decisive: composing forward-mode AD is textbook, yet it is what makes exact higher-order pushforward routine where the finite-difference alternative fails. Finally, we show no downstream task win: this is a measurement-correctness result, and whether accurate second-order geometry changes downstream conclusions is left to future work.

8 Conclusion

As geometric analysis of generative models moves from first-order bases toward curvature and geodesics, how one takes the second derivative stops being a detail. Our measurements argue for taking it with composed forward-mode AD rather than finite differences: the finite-difference second derivative of a deep generator has no usable step in single precision and no rescue in double, and its error is large enough to flip a measured sign, whereas the exact instrument costs one extra forward pass. The methods examined here stay first-order or energy-based and so avoid the problem today. Whether that holds as the field reaches for the second-order quantities the instrument now makes cheap to compute exactly is the open question this work is meant to flag.

Reproducibility statement

We release the exact-instrument module, the analytic referee, and the StyleGAN and Stable-Diffusion scripts, with fixed seeds and the emitted `metrics.json` behind every table. The generators are the public StyleGAN-bedroom-256 weights and Stable Diffusion v1.5; all experiments run on a single 8 GB GPU. The code link is withheld for anonymous review and will be added for the camera-ready version.

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A Stable-Diffusion experimental setup

The sign-inversion experiment of Section 2 uses Stable Diffusion v1.5 at 256×256 with a 20-step DDIM schedule. The h-space is the bottleneck activation of the U-Net mid-block, of shape $(1, 1280, 4, 4)$. For a base latent we run the deterministic DDIM chain to the edit step $t_{\text{edit}} = 12$, then study the map $\alpha \mapsto x_T(\alpha)$ obtained by injecting a perturbation $\alpha\, v$ into h-space at step t_{edit} and completing the chain. The prompt is fixed ("a photograph of a face"), classifier-free guidance is on, and all randomness is seeded.

We draw $K = 16$ unit-norm h-space probes, build the first-order Jacobian column by column as Je_k, take its SVD, and keep the top $k = 12$ right-singular vectors as edit directions u_i with singular values σ_i. Along each u_i we form the curvature ratio $\rho_i = |\partial^2_\alpha x_T|/|\partial_\alpha x_T|$ at $\alpha = 0$. We report Spearman(σ, ρ) over the 12 directions, for each of 6 seeds (0 through 5). The finite-difference estimate of the inner second derivative uses a central difference of the JVP at $\varepsilon \in \{0.1, 0.05, 0.01, 0.001\}$; the main text reports $\varepsilon = 0.05$ and Appendix F gives all four. Every run fits in 4.6 GB of GPU memory.

B The composed-jvp instrument

Forward-mode AD returns $\partial f(x)[v]$ alongside $f(x)$ in one pass. Nesting it gives the exact second-order pushforward with no step size:

```
def pushforward2(G, x, u):       # exact d^2/d alpha^2 G(x + alpha u) at alpha=0
    f  = lambda a: G(x + a * u)                # curve through x in direction u
    df = lambda a: jvp(f, (a,), (1.0,))[1]     # first-order pushforward, exact
    _, d2 = jvp(df, (0.0,), (1.0,))            # directional derivative of df, exact
    return d2                                  # second-order pushforward
```

Here `jvp` is `torch.func.jvp`. The inner call computes $\partial_\alpha G$; the outer call differentiates it again along the same tangent. No Jacobian or Hessian is materialized, so peak memory is that of a single forward evaluation of G, independent of the output dimension $3HW$. The k-th order pushforward follows from k nested calls. For the diffusion experiments G is the DDIM chain from t_{edit} onward; the same wrapper runs unmodified on the StyleGAN synthesis network.

C Finite-difference error: derivation

Second derivative. Taylor expansion about α gives $f(\alpha \pm \varepsilon) = f \pm \varepsilon f' + \frac{\varepsilon^2}{2} f'' \pm \frac{\varepsilon^3}{6} f''' + \frac{\varepsilon^4}{24} f^{(4)} \pm \cdots$, so the odd terms cancel in the sum and

$$\frac{f(\alpha+\varepsilon) - 2f(\alpha) + f(\alpha-\varepsilon)}{\varepsilon^2} = f''(\alpha) + \frac{\varepsilon^2}{12} f^{(4)}(\alpha) + O(\varepsilon^4).$$

Each evaluated value carries a rounding error of size $\varepsilon_{\text{mach}}|f|$, and the three-term numerator accumulates $O(\varepsilon_{\text{mach}}|f|)$, divided by ε^2. The total error is therefore $g(\varepsilon) = c_1\varepsilon^2 + c_2\varepsilon_{\text{mach}}/\varepsilon^2$ with $c_1 = |f^{(4)}|/12$ and $c_2 = O(|f|)$. Setting $g'(\varepsilon) = 2c_1\varepsilon - 2c_2\varepsilon_{\text{mach}}/\varepsilon^3 = 0$ gives $\varepsilon^\star = (c_2/c_1)^{1/4}\varepsilon_{\text{mach}}^{1/4}$ and $g(\varepsilon^\star) = 2\sqrt{c_1 c_2}\, \varepsilon_{\text{mach}}^{1/2}$. Dividing by $|f''|$ gives the relative floor $2\sqrt{|f^{(4)} f|}/|f''| \cdot \varepsilon_{\text{mach}}^{1/2}$.

First derivative. The central first difference $\big(f(\alpha+\varepsilon) - f(\alpha-\varepsilon)\big)/2\varepsilon = f' + \frac{\varepsilon^2}{6} f''' + O(\varepsilon^4)$ has truncation $O(\varepsilon^2)$ and roundoff $O(\varepsilon_{\text{mach}}|f|/\varepsilon)$, so $g(\varepsilon) = c_1\varepsilon^2 + c_2\varepsilon_{\text{mach}}/\varepsilon$, minimized at $\varepsilon^\star \sim \varepsilon_{\text{mach}}^{1/3}$ with floor $\varepsilon_{\text{mach}}^{2/3}$. The first-derivative floor ($\varepsilon_{\text{mach}}^{2/3} \approx 2 \times 10^{-5}$ in fp32) is far below the second-derivative floor ($\varepsilon_{\text{mach}}^{1/2} \approx 3 \times 10^{-4}$ benign, and orders of magnitude worse once the constant $|f^{(4)} f|/f''^2$ is large), which is the asymmetry the paper rests on.

D StyleGAN sweep: full table

Table 3 reports a subset of steps. Table 5 gives the complete fp32 sweep over 6 directions (3 attributes `indoor_lighting/wood/view` and 3 random) ×16 samples. The exact curvature is 1.17.

step δ	$\lvert D^2_\delta G\rvert$	2nd-order rel. err.	1st-order rel. err.
8	7.07×10^{-3}	74.5%	63.81%
5	1.38×10^{-2}	57.3%	54.01%
3	2.65×10^{-2}	45.2%	44.07%
2	4.32×10^{-2}	56.1%	37.12%
1.5	6.09×10^{-2}	73.6%	32.39%
1	9.92×10^{-2}	106.4%	26.22%
0.7	1.50×10^{-1}	142.0%	21.34%
0.5	2.17×10^{-1}	184.2%	17.36%
0.3	3.68×10^{-1}	266.9%	12.50%
0.2	5.56×10^{-1}	355.2%	9.37%
0.1	1.06	622.8%	5.47%
0.05	1.86	1429.3%	3.05%
0.02	4.05	6772.5%	2.44%
0.01	8.74	25023.0%	5.71%
0.005	2.36×10^{1}	93346.3%	15.04%
0.001	4.09×10^{2}	1985856.3%	139.11%

Table 5: Full fp32 StyleGAN sweep. The second-order relative error is U-shaped with a floor of 45.2% at $\delta = 3$; the first-order error has a clean minimum of 2.44% at $\delta = 0.02$, confirming the pipeline.

E Precision control: full fp32 vs fp64 sweep

Table 6 gives the per-step second-order relative error in fp32 and fp64 over 3 directions $\times 8$ samples. At large steps the two precisions agree (truncation). At small steps fp32 diverges through catastrophic cancellation while fp64 does not, yet fp64 does not converge either: it plateaus near 3000%, because the chord-scale second difference of a non-smooth (LeakyReLU) generator does not approach the pointwise pushforward. The best-step floor is 47.1% in both.

step δ	fp32 2nd-order rel. err.	fp64 2nd-order rel. err.
3	47.1%	47.1%
1	96.8%	96.8%
0.3	216.8%	216.4%
0.1	431.1%	410.2%
0.03	1467.2%	748.0%
0.01	10557.7%	1203.0%
0.003	1.09×10^{5}%	1815.0%
0.001	8.35×10^{5}%	2373.4%
10^{-4}	6.59×10^{7}%	3183.4%
10^{-5}	5.58×10^{9}%	3470.7%
10^{-6}	4.67×10^{11}%	1502.6%

Table 6: Full precision control. fp64 removes cancellation (small-step columns drop by orders of magnitude) but the floor is unchanged: no precision recovers the second-order pushforward on a real generator.

F Stable Diffusion: per-seed, per-step

Table 7 gives Spearman(σ, ρ) for every seed and every finite-difference step, with the exact column for reference. The exact estimate is negative on five of six seeds. The finite-difference estimate has no step that reproduces it: at each step the across-seed mean is near zero (-0.08 to -0.34) and the sign varies seed to seed.

seed	exact	FD@0.1	FD@0.05	FD@0.01	FD@0.001
0	−0.853	+0.287	+0.182	−0.259	+0.692
1	−0.825	−0.378	−0.552	−0.287	+0.049
2	−0.916	−0.853	−0.783	−0.902	−0.916
3	−0.944	−0.168	−0.399	−0.028	−0.538
4	−0.028	+0.238	+0.552	+0.000	+0.280
5	−0.839	+0.021	−0.070	−0.580	−0.021
mean	−0.734	−0.142	−0.178	−0.343	−0.076

Table 7: Per-seed Spearman(σ, ρ) on Stable Diffusion. No finite-difference step recovers the exact column's consistent negative sign.

G Literature audit: classification criteria

For each paper in Table 4 we asked three questions from the primary source. (i) Does it compute a second-order quantity at all (curvature, a geodesic, the metric derivative / Christoffel symbols)? (ii) If so, is that quantity obtained by an explicit $/\varepsilon^2$ finite difference, by automatic differentiation, by an analytic expression, or as the exact gradient of a discretized energy? (iii) Does a reported conclusion depend on it? A paper is "exposed" only when a second-order quantity is formed by an explicit second-order finite difference. By this test: Shao et al. (2018) discretizes the geodesic energy and its gradient is the discrete Laplacian $g_{i+1} - 2g_i + g_{i-1}$, the exact gradient of $\sum \|g_{i+1} - g_i\|^2$, with no division by ε^2, so it is not exposed; Saito & Matsubara (2025) uses first-order score differences; Arvanitidis et al. (2018) forms the metric derivative analytically and integrates the geodesic ODE numerically; Yang et al. (2018) differentiates the energy with autodiff; Park et al. (2023), Chen et al. (2018), and Kwon et al. (2023) never take a second derivative. The single exposed published practice is the first-order finite difference of a network derivative, which the asymmetry of Section 3 places on the safe side.

H Reproduction

All experiments run on a single 8 GB GPU with PyTorch 2.2.2 and `numpy<2`. The analytic referee (Table 2) is a CPU script and is deterministic. The StyleGAN sweeps use the public StyleGAN-bedroom-256 weights; the diffusion experiments use Stable Diffusion v1.5. Representative commands:

```
# StyleGAN FD-vs-exact curvature sweep (Table 3, Appendix D)
python scripts/29_shao_fd_vs_exact.py --attrs indoor_lighting wood view \
    --n-random-dirs 3 --num-samples 16

# fp32 vs fp64 precision control (Appendix E)
python scripts/30_fp64_precision_control.py --num-samples 8

# Stable-Diffusion sign inversion (Table 1, Appendix F)
python experiments/diffusion/run_park_fd_vs_exact.py --n-seeds 6 \
    --K-probes 16 --top-k 12 --steps 20 --t-edit 12 --resolution 256
```

Each table's `metrics.json` is released alongside the code. The repository link is withheld for anonymous review.